

SQL Fundamentals:

SELECT & Filtering

1. SELECT *

FROM employees;

id	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. SELECT DISTINCT department

FROM employees;

department
IT
HR
Finance
Marketing

3. SELECT first-name, last-name,

FROM employees

ORDER BY salary DESC;

first-name	last-name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4. SELECT id,
 first-name,
 last-name,
 salary
 FROM employees
 ORDER By salary DESC
 LIMIT 3;

id	first-name	last-name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. SELECT id,
 first-name,
 last-name,
 department
 FROM employees
 WHERE department = 'IT'

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6. SELECT id,
 first-name,
 last-name,
 department,
 salary
 FROM employees
 WHERE department = Finance
 AND salary > 60000;

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

7. SELECT id,
first-name,
last-name,
department
FROM employees
WHERE department = 'HR' OR
department = 'Marketing';

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. SELECT id,
first-name,
last-name,
department
FROM employees
WHERE department != 'IT';

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. SELECT id,
first-name,
last-name,
department,

FROM employees

WHERE department IN ('IT', 'HR', 'Finance');

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. SELECT id,

first-name,

last-name,

department,

salary,

city

FROM employees

WHERE department = 'IT'

AND salary > 65000

AND city = 'Johannesburg';

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg